B.Tech Degree VIII Semester Examination April 2012

EC/EB/EI 804 (C) MECHATRONICS

(2002 Scheme)

Time: 3 Hours Maximum Marks: 100

I. (a) Define and explain Mechatronics. Briefly outline the scope of Mechatronics in manufacturing. (10)

(b) Differentiate between NC and CNC. Explain the relative advantages and disadvantages of NC and CNC machine tools over conventional machine tools. (15)

OR

II. (a) With the help of suitable sketches explain the operation of open loop and closed loop NC systems. (9)

(b) Write short notes on the following terms with reference to an NC or CNC system.
(i) Resolution
(ii) System accuracy
(iii) Repeatability
(iv) Back lash (16)

III. (a) Briefly explain the working of the following feedback devices.
(i) Encoders
(ii) Resolvers
(iii) Inductosyns
(iv) Tachometers (20)

(b) The terminal input voltage of a 4 – bit weighted resistor DAC is 2V. The network resistors are 15K, 30K, 60K and 120KΩ, the feedback resistor is 40KΩ. Calculate the maximum absolute output voltage and DAC gain. (5)

OR

IV. (a) What is an interpolator? Explain its function in an NC or CNC system. (5)

(b) With the help of suitable flow charts explain the working of a Reference-Word CNC interpolator. (15)

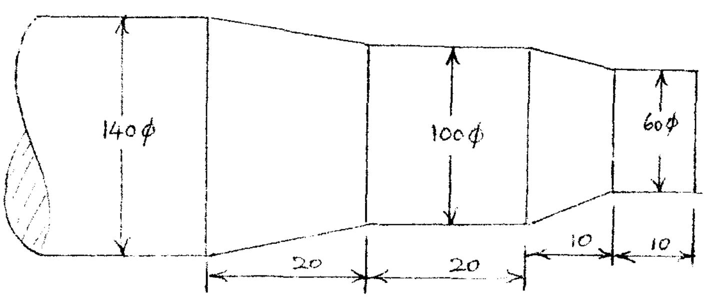
(c) Briefly explain the operation of incremental open-loop control system. (5)

V. (a) Briefly describe the various steps involved in developing an NC Manual Part Program. (5)

(b) Develop an NC Manual Part Program for machining the part given in figure. (10)

(c) Make a neat sketch of the part in figure showing dimensions, co-ordinate axes, work piece set up and tool path. (5)

(d) Explain the operations carried out in each program blocks for machining part in figure. (5)



All dimensions are in mm

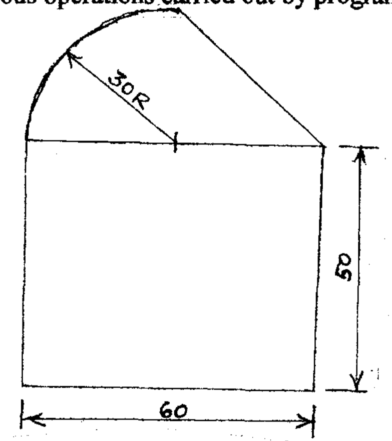
OR

VI. (a) Write short notes on:

(i) CIM

(ii) Automated storage and Retrieval system (10)

(b) Develop an APT program for the part shown in figure. Assume any missing data. Also indicate the work piece/tool set up, tool path etc. in the fully dimensioned sketch. Explain the various operations carried out by program blocks in detail. (15)



All dimensions are in mm

VII. (a) Define and explain an Industrial Robot. (5)

(b) With the help of suitable sketches explain the common Robot configurations. (20)

OR

VIII. (a) Write short notes on:

(i) Tactile array sensors

(ii) Proximity sensors (10)

(b) With the help of a block diagram explain the robot language structure. (15)

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